Adify Skills
n8n Automation Mastery

MENTOR COPY

Module 7 · Real-World Project Sprints

DAY 21 of 25

# Project Sprint 1: AI Customer Support Desk

"24×7 support jo kabhi chhutti nahi leta."

⚡ AAJ HUM YEH BANAYENGE

## AI Customer Support Desk (Chat + Email → Agent → Ticket)

Website chat aur email dono ek jagah aate hain. AI agent knowledge base se jawab deta hai; jab confident na ho ya customer "human" bole to ticket sheet mein banta hai aur team ko Slack alert jaata hai.

AI Agent | WhatsApp | Gmail | Vector Store

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

# 🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

**💡 Yeh lesson kyun zaroori hai?**

Pehla full client-grade project. Yahan student "tutorial follower" se "solution builder" banta hai.

## 🎯 Aaj ke baad aap yeh kar paoge

**01 Client requirement document (SRS) banana**
🌍 Darzi pehle naap leta hai.
⚙️ SRS: problem, users, channels, success metric, out-of-scope.

**02 Multi-channel inbox: WhatsApp + email + website chat**
🌍 Ek darwaza, kai raaste.
⚙️ WhatsApp Trigger + Gmail Trigger + Chat Trigger → common format.

**03 RAG knowledge base + AI agent**
🌍 Rule book + samajhdaar didi.
⚙️ Vector store tool + AI Agent with system prompt.

**04 Ticket creation & human escalation**
🌍 Principal ka room.
⚙️ Confidence low / "human" keyword → ticket (Sheet/Freshdesk) + Slack.

**05 CSAT feedback loop**
🌍 Smiley feedback machine.
⚙️ Reply ke baad 1-5 rating → Sheet → weekly avg.

**06 Handover documentation**
🌍 Naye manager ke liye diary.
⚙️ Loom video + README: credentials, flows, kaise edit karein.

## ⏱️ Class ka plan (2 ghante)

| 15m | 75m | 20m | 10m |
|---|---|---|---|

`0:00-0:15` **Brief & Architecture**
Client brief walkthrough
🪄 *Help desk didi ke paas rule book hai. Jo usme likha hai woh turant batati hai, baaki principal sir ko bhejti hai.*
👩‍🏫 Brief padh ke students se hi requirements nikalwao. Architecture board par sab milke banao.

`0:15-1:30` **Build Sprint**
Guided build in phases with checkpoints
🪄 *Marathon ek saath nahi daudte — har kilometer par paani peete hain. Har phase ek kilometer hai.*
👩‍🏫 Har phase ke end par 2 minute checkpoint. Jo aage hain unse peeche walon ki help karwao.

`1:30-1:50` **Demo & Review**
3 students demo
🪄 *Science fair — model dikhao, batao kaise chalta hai, sawaalon ka jawab do.*
👩‍🏫 Time strictly rakho. Har demo ke baad pehle tareef, phir ek sudhaar — tone positive.

`1:50-2:00` **Homework**
🪄 *Project ka agla page — ghar par likhna hai, kal sabko dikhana hai.*
👩‍🏫 Deadline aur submission format saaf bolo. Jo peeche hain unki list banao aur help slot do.

# Concepts — aasaan bhasha mein

🧠 Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

### SRS

Software Requirement Spec — client kya chahta hai, likhit mein.

🪄 **Analogy:** Ghar banane se pehle naksha

CONCEPT 02

### Multi-channel

WhatsApp, email, website — ek hi brain.

🪄 **Analogy:** Ek receptionist, teen phone lines

CONCEPT 03

### Escalation

AI na sambhaal paaye to human ko ticket.

🪄 **Analogy:** Junior → senior

CONCEPT 04

### CSAT

Customer satisfaction feedback loop.

🪄 **Analogy:** Restaurant ka feedback card

### 🔑 Aaj ka main idea

Real project = requirement samajhna, architecture banana, phir build. Support desk mein sab channels ek jagah aate hain, AI RAG se jawab deta hai, aur jo nahi aata woh insaan ko jaata hai.

- D2C brand ka WhatsApp support
- Coaching institute ka admission desk
- SaaS ka email support

## 📖 Ek kahani se samjho: School ka help desk

1 🙋 **Bachche har taraf se sawaal laaye**
= WhatsApp + email + chat

2 📖 **Help desk didi ne rule book dekhi**
= RAG knowledge base

3 💬 **Aasaan sawaal turant solve**
= AI agent answer

4 🎫 **Mushkil wala principal ko**
= Ticket + escalation

5 ⭐ **"Kaisa laga?" poocha**
= CSAT feedback

🪄 Ek line mein: “Help desk didi ke paas rule book hai. Jo usme likha hai woh turant batati hai, baaki principal sir ko bhejti hai.”

## 🗺️ Aaj ka workflow: AI Customer Support Desk (Chat + Email → Agent → Ticket)

Data left se right chalta hai. Har dabba ek n8n node hai.



**💰 Isse kitna kama sakte ho?**

D2C brands, coaching, SaaS (AI support desk) — aam taur par **₹50,000 – ₹1,50,000**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

## 🛠 Step-by-step build — n8n kholo aur saath chalo

1. ⚡ **WhatsApp Trigger**
   Customer message (+ Gmail Trigger for email)
2. ✏️ **Edit Fields (Set)**
   Normalize: channel, user_id, text
3. 🤖 **AI Agent**
   RAG tool + order lookup tool + Postgres Chat Memory
4. 🔀 **IF**
   needs_human = true?
5. ✅ **ClickUp**
   Ticket create + team notify
6. 💬 **WhatsApp Business Cloud**
   Reply + CSAT 👍/👎

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

### 🔑 Setup (import ke baad)

- [ ] Connect: **Gmail (Google OAuth)**
- [ ] Connect: **OpenAI API key**
- [ ] Connect: **Google Sheets (Google OAuth)**
- [ ] Connect: **Slack (OAuth / bot token)**
- [ ] Day 16 wala "Docs Search Tool" (vector store) copy karke is agent mein ai_tool ki tarah jodo
- [ ] "Tickets" tab headers: channel, user, text, output
- [ ] Slack mein #support channel

### ✅ Test kaise karein

1. Chat mein normal sawaal → AI jawab
2. "Mujhe insaan se baat karni hai" → ticket + Slack
3. Support email bhejo → reply thread mein aaya?

## ⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

**😫 Pehle (manual)**

3 log, 3 inbox, har reply manually — **480 min**

480 min

**⚡ Ab (n8n)**

70% AI se, 30% team ko ticket — **120 min**

120 min

⏱ Support team ka ~75% time bacha, reply time ghanton se seconds.

## ⌨️ Aaj ka key snippet

**Copy karke n8n mein paste karo** 📋 Select karke copy karo

```
Escalate (needs_human=true) if: refund, legal, angry tone, or 2 failed answers.
```

## 📦 Sample data — input se output tak

**📥 Input** 📋 Select karke copy karo

```
{
  "channel": "whatsapp",
  "from": "+919811122233",
  "text": "Order #4521 kab aayega?"
}
```

**📤 Output** 📋 Select karke copy karo

```
{
  "reply": "Aapka order #4521 kal tak deliver ho
jayega 🚚",
  "needs_human": false,
  "csat": "asked"
}
```

## 💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

**💬 Support agent system prompt** 📋 Select karke copy karo

```
Tum "YourBrand" ke support agent ho. Policies:
- Refund 7 din ke andar, unused product.
- Delivery 3-5 working days.
- Order status ke liye order ID maango.
Sirf in policies aur knowledge tool se jawab do. Agar customer gussa hai, refund ₹5000 se upar hai, legal
baat hai, ya tumhe pakka nahi pata — sirf likho: ESCALATE: <ek line reason>
```

**💬 Client SRS template prompt** 📋 Select karke copy karo

```
Act as an automation consultant. Client: {{business}}. Problem: {{problem}}.
Write a 1-page SRS with: goal, users, channels, current process (steps + time), proposed workflow
(trigger → steps → output), integrations & credentials needed, edge cases, success metrics, out of scope,
timeline. Keep it simple and in bullet points.
```

**💬 Build it with ChatGPT / Claude** 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Website chat aur email dono ek jagah aate hain. AI agent knowledge base se jawab deta hai; jab
confident na ho ya customer "human" bole to ticket sheet mein banta hai aur team ko Slack alert jaata
hai.

Flow: Chat + Gmail (2 channels) → Edit Fields (Common format) → AI Agent (KB se jawab) → IF (ESCALATE?) →
Ticket + Slack / Reply (Insaan ya AI)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to
test it and the 3 most common errors beginners hit.
```

💬 **Debug helper prompt** 📋 Select karke copy karo

```
I'm building the n8n workflow "AI Customer Support Desk (Chat + Email → Agent → Ticket)" (nodes: Website
Chat, Support Email, Normalize Chat, Normalize Email, Support Agent, Agent Model, Agent Memory,
Escalate?, Create Ticket, Alert Team, Email or Chat?, Reply Email, Chat Answer).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

## 📥 Workflow ko n8n mein kaise laayein

1. **QR scan karo** ya link kholo — workflow ki JSON file khulegi.
2. Poora text copy karo ( `Ctrl` + `A` , `Ctrl` + `C` ).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ **Workflow JSON**
13 nodes · ready to import
`adifyskills.com/workflows/day-21-ai-support-desk.json`

🌐 **Lesson page**
Copy buttons, quiz aur video links
`adifyskills.com/day/21`

# Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

| Agar yeh ho raha hai… | To yeh karo |
|---|---|
| **Node par laal ⚠️ triangle?** | Account connect nahi hua. Node kholo → Credential → "Gmail (Google OAuth)" connect karo. |
| **Node par laal ⚠️ triangle?** | Account connect nahi hua. Node kholo → Credential → "OpenAI API key" connect karo. |
| **Node par laal ⚠️ triangle?** | Account connect nahi hua. Node kholo → Credential → "Google Sheets (Google OAuth)" connect karo. |
| **Node par laal ⚠️ triangle?** | Account connect nahi hua. Node kholo → Credential → "Slack (OAuth / bot token)" connect karo. |
| **Webhook/Form test par kuch nahi ho raha?** | "Listen for test event" / Test URL use karo aur 2 minute ke andar request bhejo. Production URL sirf Active workflow par chalta hai. |
| **Time par apne aap nahi chala?** | "Active" toggle on hai? Settings mein timezone Asia/Kolkata check karo. |
| **AI node par 401 / 429 error?** | 401 = API key galat. 429 = credit khatam ya bahut jaldi requests. OpenAI billing page check karo. |
| **"Sheet not found" / column khaali?** | Sheet ID aur tab ka naam (capital/small bhi) check karo. Pehli row mein headers hone chahiye. |

### ⚠️ Yeh galtiyan sabse common hain

- ! Har channel ke liye alag agent (maintenance nightmare)
- ! Escalation rules vague
- ! Handover docs ke bina deliver

### ⭐ Pro tips — time bachane wali tricks

- ★ Ek normalized "message" format banao — sab channels usme convert
- ★ Har AI reply log karo — weekly review
- ★ 3 min Loom demo = portfolio gold

**Phir bhi atke? Koi baat nahi 🙂**
Screenshot + error message ke saath WhatsApp par poochho →

# Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

**🛠 Live lab**

Full AI support desk: WhatsApp + email → AI agent (RAG) → answer ya ticket + escalation.

**📝 Homework**

Project ko polish karo, Loom demo video (3 min) record karo.

- [ ] Homework ho gaya

**BEGINNER**

Sirf email channel

- [ ]

**INTERMEDIATE**

WhatsApp + email

- [ ]

**ADVANCED**

Website chat + analytics dashboard

- [ ]

### 🧩 Aaj ka challenge — pehle khud try karo

**🎯 Har ticket ke saath ek ticket number (TKT-1234) banao aur customer ko batao.**

**💡 Hint 1**

Create Ticket se pehle ek Set node mein number banao.

**💡 Hint 2**

Expression: {{ 'TKT-' + $now.toFormat('MMddHHmm') }}

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

# 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

**Q1. Project shuru karne se pehle?**

- (A) Logo
- (B) Invoice
- (C) Deploy
- (D) SRS / requirements

**Q2. AI na sambhaal paaye to?**

- (A) Human escalation
- (B) Ignore
- (C) Delete
- (D) Retry forever

**Q3. Multi-user memory ke liye best?**

- (A) Sheet cell
- (B) Postgres Chat Memory
- (C) No memory
- (D) Sticky note

**Q4. Customer satisfaction metric?**

- (A) SQL
- (B) API
- (C) CSAT
- (D) CPU

## 📖 Aaj ke naye shabd

**Trigger**
Jo workflow ko START karta hai — jaise alarm, naya email, ya form submit.

**Agent**
AI jo khud decide karta hai kaunsa tool use karna hai — sirf baat nahi, kaam bhi karta hai.

**RAG**
AI ko pehle aapke documents mein se sahi hissa dhoondh ke dena, phir jawab banwana — open-book exam.

# 👨‍🏫 Mentor notes — class kaise lein

Yeh page sirf aapki copy mein hai. Students ki PDF mein nahi jaata.

- [ ] Client brief role-play se shuru
- [ ] Build ko 4 checkpoints mein baanto, har checkpoint par sync
- [ ] 3 students live demo

## 🗣️ Har block ke liye mentor tip

**15m — Brief & Architecture**

Brief padh ke students se hi requirements nikalwao. Architecture board par sab milke banao.

Class se poochho: SRS mein kya-kya likhte hain? · Escalation kab honi chahiye? · CSAT kaise measure karein?

**75m — Build Sprint**

Har phase ke end par 2 minute checkpoint. Jo aage hain unse peeche walon ki help karwao.

Class se poochho: Kaunsa phase complete hua? · Kahan error aa raha hai? · Agle 15 minute ka target kya hai?

**20m — Demo & Review**

Time strictly rakho. Har demo ke baad pehle tareef, phir ek sudhaar — tone positive.

Class se poochho: Is workflow ne kitna time/paisa bachaya? · Error aaya to kya hoga? · Isko client ko kitne mein bechoge?

**10m — Homework**

Deadline aur submission format saaf bolo. Jo peeche hain unki list banao aur help slot do.

Class se poochho: Aapka capstone problem statement kya hai? · Kis step par help chahiye? · Kal tak kya complete karoge?

## ✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 22

**Project Sprint 2: AI Content & Marketing Engine**

"Ek idea → 30 din ka content."

☐ **Day 21 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

# 🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

**Q1 → D**
SRS / requirements

**Q2 → A**
Human escalation

**Q3 → B**
Postgres Chat Memory

**Q4 → C**
CSAT

### ✅ Challenge ka solution

Escalate? true → Set ticket_no = {{ 'TKT-' + $now.toFormat('MMddHHmm') }} → Create Ticket → reply mein ticket_no bhejo.

## 📚 Official docs

- Support agent templates ↗
- WhatsApp node ↗
- Human in the loop ↗
- Postgres Chat Memory ↗

## ▶ Videos

- n8n ai customer support agent ↗
- n8n whatsapp ai agent rag ↗

### 💬 Doubt poochho

WhatsApp par seedha message

`wa.me/919000000000?text=Day%2021%20(Project%20Sprint%201%3A%20AI%20Customer%20Support%20Desk)%20mein%20doubt%20hai%3A%20`

### 🌐 Day 21 lesson page

Quiz, copy buttons, videos

`adifyskills.com/day/21`